

French Motor Insurance: Frequency, Severity, and Expected Claim Cost

Nathan Mehlhop

September 29, 2026

Abstract

We estimate annual expected claim cost for French motor third-party liability policies by modeling claim frequency and mean claim severity separately. After reconciling 677991 policy records with 26444 claim records, we compared candidate models using five policy-level folds within a development sample and evaluated the selected specifications on reserved policies. A boosted Poisson frequency model and a Gamma log-link severity model predicted 11.663 million in test loss against 11.734 million observed, an observed-to-predicted ratio of 1.006. Frequency performance was consistent between development and test data. Severity remained less reliable across risk groups and sensitive to rare large claims. Comparison with more flexible and explicit tail models informed a judgment to retain the comparatively stable severity mean model while qualifying what aggregate calibration can establish.

Contents

1	Introduction	1
2	Data and Preparation	2
3	Exploratory Findings	3
4	The Pricing Model	4
5	Evaluation on Reserved Policies	5
5.1	Portfolio Expected Loss	5
5.2	The Two Model Components	6
6	Model Selection and Alternatives	7
6.1	Development Cross-Validation	7
6.2	Lessons from More Flexible Severity Models	9
7	Conclusion	10

1 Introduction

An insurer needs an estimate of expected claim cost for each policy, accounting for both how often claims occur and how much they cost when they occur. This project uses driver, vehicle, and geographic characteristics to estimate those two quantities separately. If $\lambda(X)$ is expected claims

per exposure-year for a policy with characteristics X , and $\mu(X)$ is expected amount per claim, then its annual expected claim cost, or pure premium, is

$$PP(X) = \lambda(X)\mu(X).$$

For a policy observed for e years, its expected loss over that period is $e PP(X)$. Pure premium represents expected claim cost; it does not include expenses, profit provisions, or other components of a charged premium.

The main challenge is the difference between the two model components. Claim frequency shows fairly clear patterns across policy characteristics. Claim amounts have a long right tail, and a small number of losses can materially change the arithmetic mean needed for pricing. We used separate exploratory and modeling analyses to understand these features, compared candidates on development data, and evaluated the resulting pricing model once on reserved policies.

The final model combines boosted Poisson frequency with Gamma GLM severity, producing an aggregate prediction within 0.6% of observed loss on the reserved policies. The closer inspection that led to this choice is as important as the headline result. Frequency patterns were sufficiently stable to support a flexible model, while severity estimates were sensitive to a few large claims, weak across risk groups, and unstable under an explicit body/tail specification. We therefore selected the severity model for the relative stability of its conditional-mean predictions, not because a Gamma distribution fully describes the claim tail. This distinction matters for pricing: good aggregate agreement is useful, but it does not establish reliable individual risk ranking or protection against unusually large losses.

The accompanying Python notebooks provide the full analysis behind this report. Notebook 01 documents data intake and integrity checks. Notebooks 02 and 03 explore claim frequency and severity, respectively. Notebook 04 records the initial GLM specifications and the policy-level split, including an extrapolation problem that informed later model choices. Notebook 05 compares candidate frequency and severity models using development-sample cross-validation, and Notebook 06 investigates whether an explicit body/tail model improves severity estimates. Notebook 07 refits the selected frequency and severity specifications on the development sample and evaluates them on the reserved policies. Together, the notebooks provide the supporting tables, diagnostic plots, and implementation details for the results summarized here.

2 Data and Preparation

The analysis uses `freMTPL2freq` and `freMTPL2sev` from the `CASdatasets` R package, exported to CSV and analyzed in Python. The frequency file contains 677991 policy exposure periods, with claim count, exposure in years, and driver, vehicle, and geographic attributes. The severity file contains 26444 claim amounts. Multiple claims may belong to one policy.

Before combining the files, we checked missing values, policy identifiers, exposure and outcome ranges, and claim-count consistency. Neither file has missing values. Policy identifiers are unique in the frequency file, and every severity record matches a policy. The sum of the policy claim counts is 26444, exactly equal to the number of severity records; the counts also agree for every individual policy. Identifiers were read as text to preserve their stored representation. All exposures and claim amounts are positive, and claim counts are nonnegative. The policy table and claim table have different units of observation. Frequency calculations use policy claim counts divided by exposure. Severity calculations use claim records, so a policy with multiple claims contributes multiple amounts.

Initial descriptive summaries used the full supplied portfolio before the policy split was established. For model development, policies were assigned to an 80% development sample and a reserved 20% test sample; all claims associated with a policy followed its assignment. Candidate fitting and comparison used five policy-level folds within the development sample. The reserved policies were excluded from model fitting and candidate selection, but they were not wholly unseen during exploratory analysis. The final test metrics should therefore be interpreted as an out-of-sample assessment for fitting and selection, subject to this qualification. The development sample contains 542830 policies and 21079 claims; the final test sample contains 135161 policies and 5365 claims.

3 Exploratory Findings

The full portfolio contains 26444 claims over 358482.84 exposure-years, or 7.38 claims per 100 exposure-years. A group frequency is calculated as total claims divided by total exposure within that group. Averaging policy-level claim-to-exposure ratios would give very short exposures disproportionate influence.

Driver age and bonus-malus show particularly strong marginal associations with frequency. Drivers aged 18–24 have 16.10 claims per 100 exposure-years, more than twice the portfolio rate. Examining individual ages shows a decline throughout the twenties rather than a single change at age 25 (Figure 1). Frequency also rises substantially across bonus-malus groups. These comparisons describe observed groups; they do not isolate causal effects or adjust for other policy characteristics.

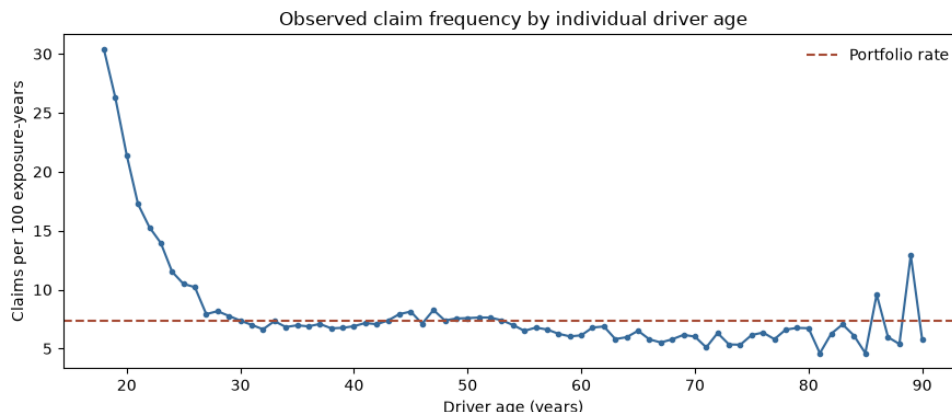


Figure 1: Observed claims per 100 exposure-years by individual driver age. Ages with fewer than 100 exposure-years are omitted from the plot.

Geography provides another descriptive pattern. Frequency rises from 5.43 claims per 100 exposure-years in rural Area A to approximately 9.5 in the most urban Area categories. Population-density groups show a similar progression. Area and density capture overlapping information, so their separate marginal patterns do not establish independent effects.

Claim severity presents a different problem. The full claim file has a mean amount of 2265.51 and a median of 1172.00; its largest claim exceeds 4.07 million. The largest 1% of claims account for 38.0% of total recorded amount, and the largest 5% account for 52.1% (Figure 2). Several amounts occur unusually often, consistent with the settlement conventions noted in the dataset documentation.

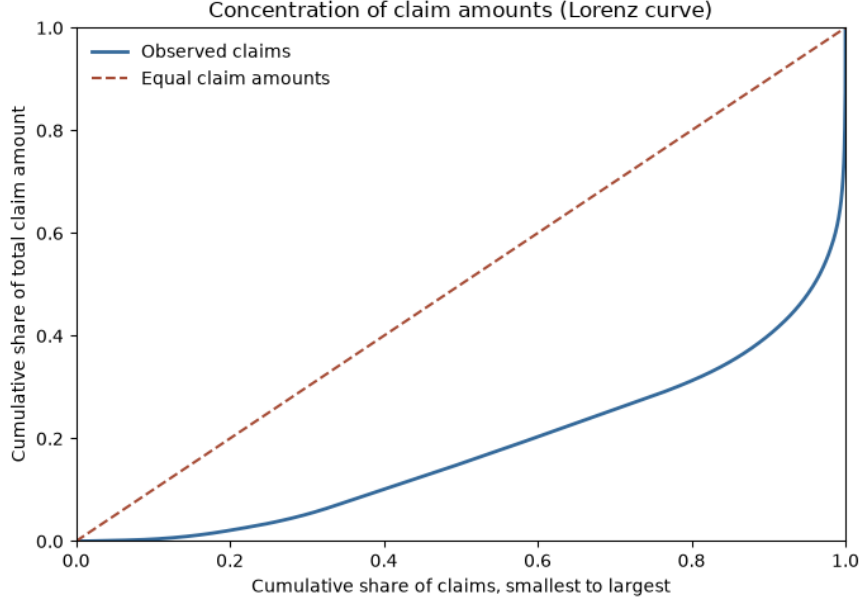


Figure 2: Cumulative share of claim amount against cumulative share of claims, ordered from smallest to largest. The dashed diagonal represents equal claim amounts.

Observed mean severity varies across some policy groups, but many group medians remain close to 1172. As a sensitivity check, setting amounts above the portfolio-wide 99th percentile to that percentile reduces the overall mean from 2265.51 to 1569.16. Several apparently large group differences also shrink. This calculation diagnoses the influence of the upper tail; the large claims remain in the modeling target. Because pricing requires an arithmetic mean, performance on ordinary claims alone cannot determine which severity model is preferable.

4 The Pricing Model

The selected frequency model is a histogram gradient boosting regressor with a Poisson loss. It uses driver age, bonus-malus, vehicle age and power, fuel type, population density, and region. The fitted target is claims per exposure-year, with policy exposure used as a weight. Its prediction $\hat{\lambda}(X)$ is therefore an annual expected frequency; multiplying by exposure gives the policy's predicted claim count.

The severity model is a Gamma generalized linear model with a log link. It uses log driver age, log bonus-malus, $\log(1 + \text{vehicle age})$, vehicle power, fuel type, log density, and region. It is fitted to positive claim amounts and predicts the arithmetic conditional mean

$$\hat{\mu}(X) \approx E[Y \mid \text{a claim occurs}, X].$$

The log link keeps this mean positive. Use of a Gamma GLM for the conditional mean does not require treating its fitted distribution as a complete description of individual claims, particularly in the extreme tail.

For each policy, we combine the two fitted means:

$$\widehat{\text{PP}}(X) = \hat{\lambda}(X)\hat{\mu}(X), \quad \widehat{L}(e, X) = e \widehat{\text{PP}}(X).$$

The severity prediction is evaluated for every policy, including those with no observed claim. The predicted portfolio loss is the sum of $\hat{L}(e, X)$ over policies, compared with the sum of their recorded claim amounts.

These specifications were chosen through development-sample cross-validation and fixed before final evaluation. The comparison with other candidates, including an explicit extreme-loss model, is discussed after the test results.

5 Evaluation on Reserved Policies

5.1 Portfolio Expected Loss

The 135161 test policies contribute 71379.17 exposure-years and 11.734 million in observed claim amounts. The combined model predicts 11.663 million in loss. The aggregate observed-to-predicted ratio is 1.006; expressed annually, observed loss is 164.39 per exposure-year versus a predicted pure premium of 163.40.

This is close portfolio-level calibration for the reserved sample. It does not imply that individual policy losses are predictable or that every risk group is equally well calibrated. Figure 3 compares observed and predicted loss rates across deciles of predicted annual pure premium. The model differentiates expected cost substantially, but realized losses fluctuate around its predictions: the fourth decile has an observed-to-predicted ratio of 1.70, while the ninth has a ratio of 0.78.

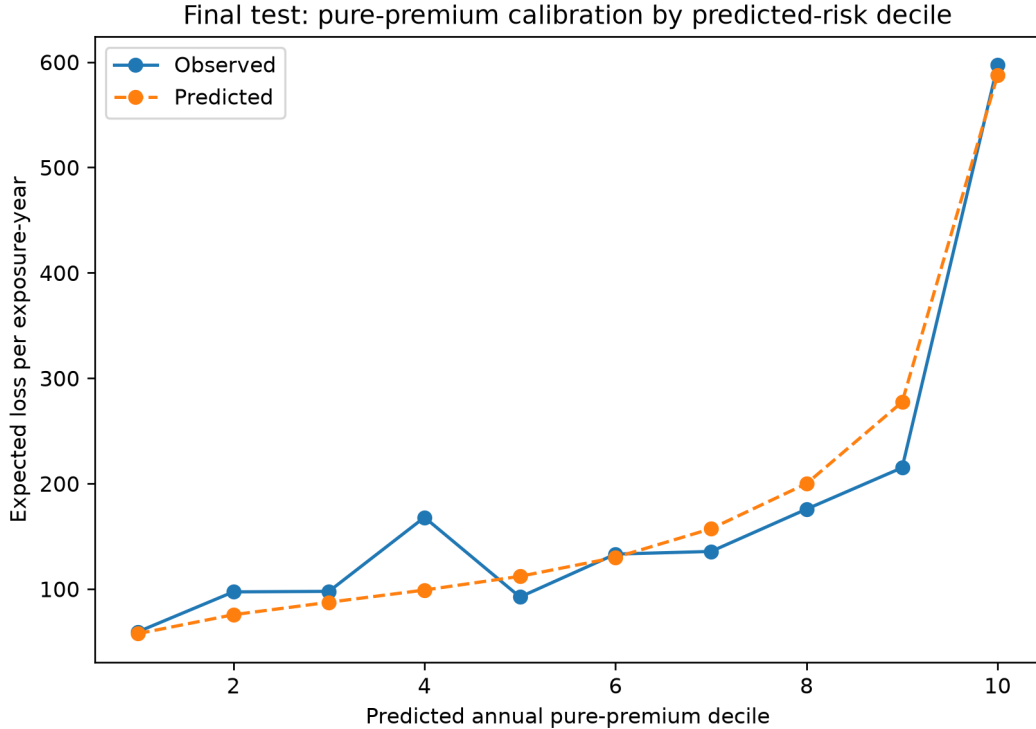


Figure 3: Observed and predicted annual pure premium across deciles formed from the frozen model’s predictions on reserved policies.

Across these pure-premium deciles, predicted annual frequency rises from 0.038 to 0.198, while predicted conditional severity rises from 1620 to 3116. Both components contribute to the ranking

of expected cost. Frequency supplies the larger and more reproducible differentiation.

5.2 The Two Model Components

The frequency model predicts 5232 claims against 5365 observed on test, giving an observed-to-predicted ratio of 1.025. Its mean Poisson deviance is 0.23869, close to the development out-of-fold value of 0.23716. Figure 4 shows generally similar observed and predicted annual rates across frequency deciles, although the highest predicted-frequency decile is underpredicted by approximately 11%.

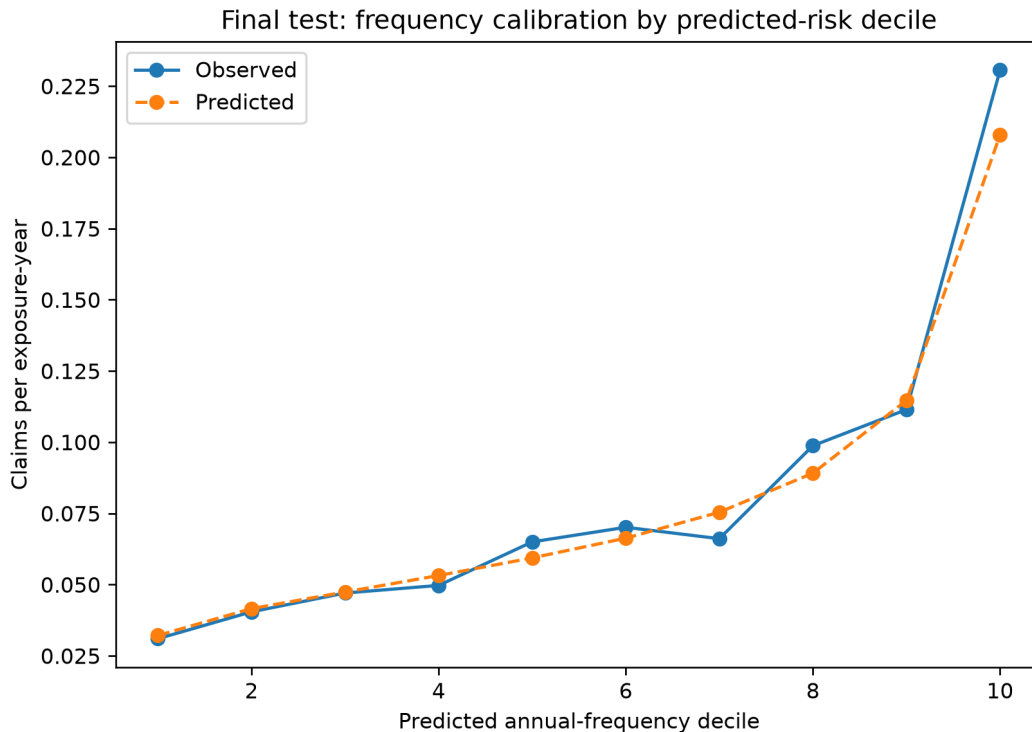


Figure 4: Observed and predicted annual claim frequency across test deciles formed from predicted annual frequency.

Among the 5365 test claims, the Gamma model predicts a mean amount of 2218.59 against an observed mean of 2187.15, for an aggregate observed-to-predicted ratio of 0.986. Calibration across predicted-severity quintiles is considerably weaker. In the highest quintile, the predicted mean is 3394 while the observed mean is approximately 2100 (Figure 5). Observed quintile means do not increase consistently with predicted means. Close aggregate severity calibration therefore does not establish that the model accurately ranks conditional claim amounts.

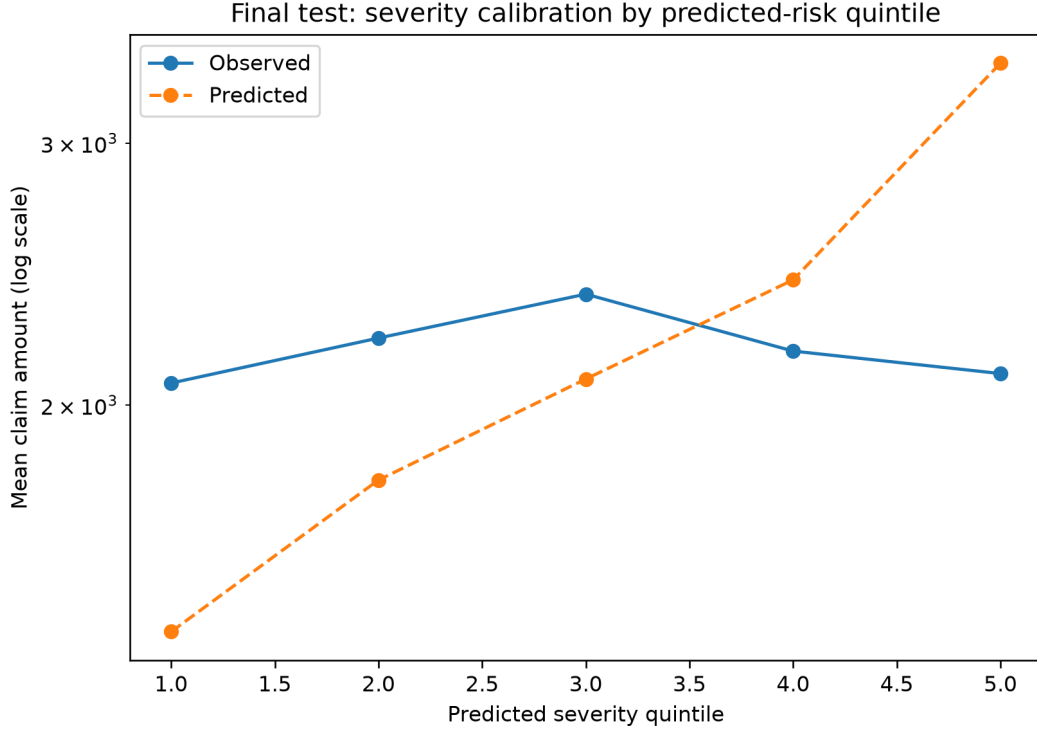


Figure 5: Observed and predicted mean claim amounts across quintiles of predicted severity on test claims.

The test data independently reproduce the concentration seen during exploration: the largest 1% of test claims account for 36.1% of total test loss, and the largest 5% account for 50.8%. If the 20 largest observed test claims are excluded solely when recalculating diagnostics, with the fitted model and predictions unchanged, severity observed-to-predicted falls from 0.986 to 0.701. Those claims are included in the actual loss target and evaluation. Their influence explains why portfolio-level severity agreement remains sensitive to the particular large losses realized in a sample.

6 Model Selection and Alternatives

6.1 Development Cross-Validation

Every candidate was evaluated with the same five policy-level development folds. Each development policy, and each claim on that policy, received a prediction from a model fitted without its fold. Frequency models were compared using Poisson deviance on policy claim counts and calibration by predicted annual frequency. Severity models were compared on the original claim-amount scale, with particular attention to aggregate loss, risk-group calibration, fold stability, and tail sensitivity. The test set played no role in these comparisons.

The Poisson and Negative Binomial GLMs had nearly identical mean out-of-fold Poisson deviance of 0.24305. Boosted Poisson reduced it to 0.23716, an improvement of approximately 2.4% that occurred in every fold. All three predicted approximately the correct total number of claims, but boosted Poisson was generally better calibrated across its predicted-frequency deciles (Figure 6). The Negative Binomial specification did not improve conditional-mean prediction despite overdispersion in the policy counts.

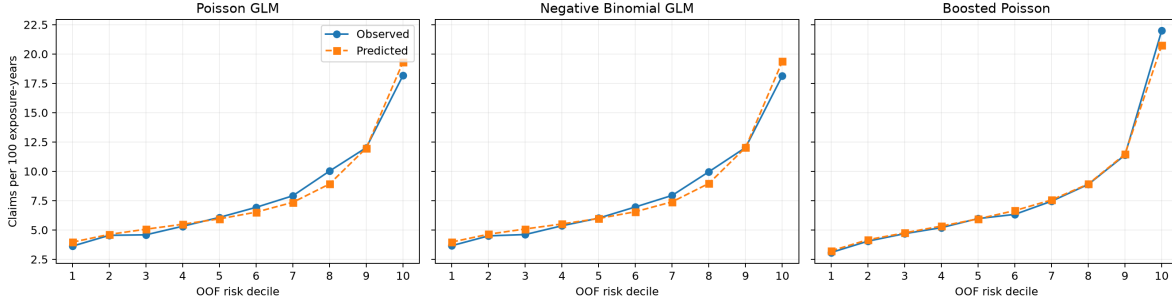


Figure 6: Development out-of-fold calibration across deciles of predicted annual frequency. Each candidate defines its own deciles.

Severity selection was less clear. Table 1 summarizes the principal conventional candidates. Boosted Gamma and normally retransformed lognormal regression lowered mean absolute error relative to Gamma, but substantially underpredicted the aggregate arithmetic mean. Empirical smearing of the same log regression brought its pooled mean closer to the observed mean, while its correction factor and risk-group calibration remained sensitive to large claims. RMSE was dominated by catastrophic losses and differed little among candidates.

Table 1: Out-of-fold severity performance on the development claims. The observed mean is 2285.46; O/P is total observed divided by total predicted amount.

Model	Predicted Mean	O/P	MAE
Gamma GLM	2228	1.026	2289
Boosted Gamma	1799	1.271	1981
Lognormal, normal correction	1778	1.286	1946
Lognormal, smearing correction	2319	0.985	2333

The lognormal comparison illustrates why fitting log amounts is not itself enough for pricing. If a model estimates $m(X) = E[\log Y | X]$, then $\exp(m(X))$ is not generally $E[Y | X]$. The normal-theory and empirical-smearing corrections applied to the same fitted log regression produced materially different original-amount predictions. In one fitting fold, the largest exponential residual contributed more than 10% of the smearing sum.

Figure 7 shows a further difficulty. Excluding only the largest observed claim in each validation fold for diagnostic evaluation removes five claims but 13.0% of observed development loss. The Gamma GLM’s pooled observed-to-predicted ratio moves from 1.026 to 0.893; excluding the largest 20 per fold moves it to 0.694. Predictions and fitting data are unchanged in this calculation. A lower error on the numerous moderate claims cannot, by itself, establish a better estimate of full expected loss.

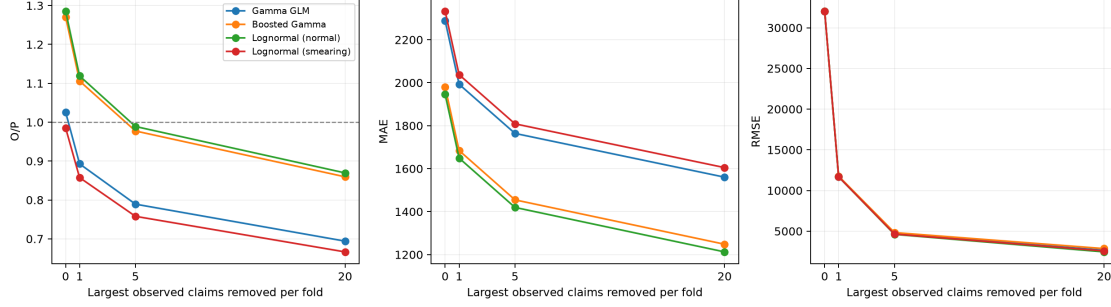


Figure 7: Out-of-fold severity metrics after diagnostic exclusion of the largest observed claims within each validation fold. Fitted models and predictions remain fixed.

6.2 Lessons from More Flexible Severity Models

An initial Poisson and Gamma GLM specification used unrestricted spline terms for age and bonus-malus. It provided a useful first adjusted comparison, but one severity validation claim had a bonus-malus value of 228 while that fold’s fitting claims reached only 190. The spline and log link produced a predicted amount above two billion for a claim of 1128.12. The conventional models in the systematic comparison therefore used simpler transformed predictors, and fitted responses were checked across folds. This episode was a model-support problem rather than an extreme observed loss.

We also tested an explicit body/tail severity model. For a threshold u , it estimated a truncated Gamma body mean $\mu_B(X)$, an exceedance probability $p(X) = P(Y > u \mid X)$, and a generalized Pareto distribution (GPD) for the excess above u . With GPD scale β and shape ξ , its arithmetic mean prediction was

$$E[Y \mid X] = (1 - p(X))\mu_B(X) + p(X) \left(u + \frac{\beta}{1 - \xi} \right), \quad \xi < 1.$$

This decomposition distinguishes the chance of an extreme claim from its expected amount. The Gamma body was normalized for truncation at u ; the GPD scale and shape were held constant across policies.

Thresholds of 5000, 10000, and 20000 produced materially different predicted means. At $u = 10000$, one fitting fold estimated $\xi = 0.9882$, making $1 - \xi$ small and yielding a conditional tail mean near 744000. Omitting just the largest fitting claim as a diagnostic reduced that estimate to approximately 142000. In policy-level bootstrap resamples of that fold, 40.5% of GPD fits had $\xi \geq 1$, for which the fitted distribution has no finite mean. Figure 8 shows the uncertainty in shape across folds and thresholds.

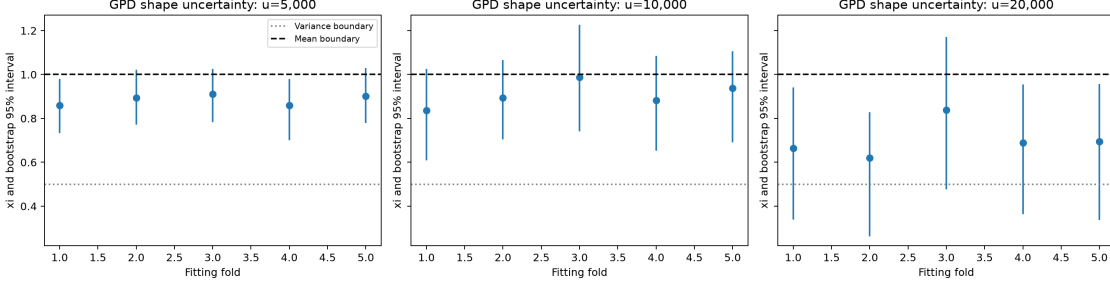


Figure 8: Policy-level bootstrap variation in fitted GPD shape across thresholds and development folds.

A body/tail candidate at $u = 20000$ achieved slightly closer pooled aggregate calibration than Gamma, but had more than twice the fold variation in predicted mean and slightly worse mean absolute and root mean squared errors. Lower thresholds substantially overpredicted aggregate loss. The component fits passed numerical checks; the limiting issue was information about the extreme tail, not failed optimization. Explicitly fitting a heavy-tailed distribution did not supply a more dependable estimate of its arithmetic mean. We therefore retained the Gamma GLM as the comparatively stable conditional-mean severity model among the approaches tested, without treating it as a correct complete distribution for claim amounts.

7 Conclusion

The selected frequency–severity model predicted 11.663 million in loss against 11.734 million observed on reserved policies, an aggregate observed-to-predicted ratio of 1.006. Boosted Poisson frequency performance closely matched development cross-validation and distinguished policies with substantially different claim rates. The combined model also differentiated expected cost across policy groups.

The main qualification is severity. Its aggregate test mean was close to observed, but calibration across predicted-severity groups was weak, and rare large losses had substantial influence in both development and test samples. More flexible severity alternatives and an explicit body/GPD-tail model clarified this limitation but did not provide a more stable estimate of expected claim amount. We retained the Gamma GLM as the comparatively stable conditional-mean model among those tested, without treating it as a complete description of the claim tail. The results support its use for portfolio and grouped expected-loss estimation in this dataset, while limiting claims about individual severity ranking and catastrophic-loss prediction.